Given a table named ‘events’ with a column ‘items’ that contains an array of values, which SQL query would transform each item of the array into a separate row?

A SELECT user_id, event_timestamp, event_name, ARRAY(items) AS item FROM events;

B SELECT user_id, event_timestamp, event_name, items FROM events;

C SELECT user_id, event_timestamp, event_name, UNNEST(items) AS item FROM events;

D SELECT user_id, event_timestamp, event_name, explode(items) AS item FROM events;

Question 17

A data engineer is overwriting data in a table by deleting the table and recreating the table. Another data engineer suggests that this is inefficient and the table should simply be overwritten instead. Which of the following reasons to overwrite the table instead of deleting and recreating the table is incorrect?

A. Overwriting a table is efficient because no files need to be deleted.

B. Overwriting a table results in a clean table history for logging and audit purposes.

C. Overwriting a table maintains the old version of the table for Time Travel.

D. Overwriting a table is an atomic operation and will not leave the table in an unfinished state.

E. Overwriting a table allows for concurrent queries to be completed while in progress.

An engineering manager wants to monitor the performance of a recent project using a Databricks SQL query. For the first week following the project's release, the manager wants the query results to be updated every minute. However, the manager is concerned that the compute resources used for the query will be left running and cost the organization a lot of money beyond the first week of the project's release.
Which approach can the engineering team use to ensure the query does not cost the organization any money beyond the first week of the project's release?

A. They can set a limit to the number of DBUs that are consumed by the SQL Endpoint.

B. They can set the query's refresh schedule to end after a certain number of refreshes.

C. They can set the query's refresh schedule to end on a certain date in the query scheduler.

D. They can set a limit to the number of individuals that are able to manage the query's refresh schedule.

A data engineer wants to create a relational object by pulling data from two tables. The relational object does not need to be used by other data engineers in other sessions. In order to save on storage costs, the data engineer wants to avoid copying and storing physical data.
Which of the following relational objects should the data engineer create?

A. Spark SQL Table

B. View

C. Database

D. Temporary view

E. Delta Table

In a scenario where you are tasked with integrating data from multiple sources into a single target table within a Databricks environment, you need to ensure the solution is efficient, handles duplicates, and manages deletions appropriately. Considering the constraints of cost, compliance, and scalability, which of the following commands is the BEST choice for this task and why? Choose the most appropriate option.

A. CREATE OR REPLACE TABLE, because it simplifies the process by allowing you to either create a new table or replace an existing one with the combined data, but it does not efficiently handle duplicates or deletions.

B. INSERT OVERWRITE, because it efficiently overwrites the target table with new data from multiple sources, but it lacks the capability to handle duplicates and deletions properly.

C. MERGE, because it not only combines data from multiple sources into the target table but also efficiently handles duplicates and deletions, ensuring data integrity and compliance.

D. COPY INTO, because it is designed for inserting data from multiple sources into a target table while preventing duplication, but it does not manage deletions or updates as effectively as other methods.

In the context of processing a large DataFrame 'df' in a Databricks environment, which contains a column 'array_column' with arrays of integers, you are tasked with transforming the data to facilitate detailed analysis of each individual element within these arrays. The transformation must support operations like filtering, sorting, and aggregation on each element separately. Considering the constraints of maintaining data integrity, optimizing for performance in a cloud environment, and ensuring the solution is scalable for datasets expected to grow significantly over time, which of the following approaches is most appropriate? Additionally, the solution should minimize the need for custom code to reduce maintenance overhead. Choose the best single option.

A	Utilize the 'explode' function to generate a new row for each element in the array, enabling individual analysis of each element. This method leverages built-in Spark functionality for optimal performance and scalability.
B	Implement the 'flatten' function to merge all elements into a single string, simplifying the dataset for bulk operations. While this reduces row count, it compromises the ability to analyze individual elements directly.
C	Develop a custom UDF (User Defined Function) to pre-filter elements based on specific conditions before applying the 'explode' function. This offers tailored flexibility but at the cost of increased complexity and potential performance drawbacks.
D	First apply 'flatten' to condense the array into a single string, then use 'explode' to separate each character for analysis. This approach is inefficient and unsuitable for analyzing individual integer elements as required.
E	Both 'A' and 'C' are viable under different circumstances. 'A' is preferable for general use due to its simplicity and performance, whereas 'C' is better suited for scenarios requiring preliminary filtering.

Your organization is moving from a traditional data lake to a data lakehouse architecture to enhance its data analytics capabilities. What is a key advantage of this transition?

A To reduce the amount of data stored

B To eliminate the need for data processing

C To enhance data quality and governance with schema enforcement

D To increase the complexity of data queries

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To effectively monitor the performance and status of Delta Live Tables (DLT) pipelines, which method provides the ability to query the event log for detailed metrics and update history?

A. Directly accessing the DLT UI for real-time metrics without the need for querying.

B. Querying the Delta Lake table directly for pipeline metrics, bypassing the event log.

C. Executing the command `SELECT * FROM event_log` to fetch all entries from the event log.

D. Establishing a temporary view on the event log path and querying this view to obtain specific metrics.

In the context of data governance within an Azure Databricks environment, data classification plays a pivotal role in managing and protecting data effectively. Considering a scenario where an organization is required to comply with GDPR and other regulatory requirements, which of the following best describes the concept of data classification and its application to enhance data governance? Choose the best option from the four provided.

A. Data classification is the process of organizing data solely based on its physical storage location, such as on-premises or cloud, to optimize retrieval times.

B. Data classification involves categorizing data based on the frequency of access, such as hot, cool, or archive, to reduce storage costs.

C. Data classification is the process of labeling data based on its sensitivity and criticality, such as public, confidential, or restricted, to implement appropriate security measures and compliance policies.

D. Data classification is the method of sorting data by its creation date to ensure older data is archived or deleted according to retention policies.

A data engineer has created an orders Delta table. A junior data engineer, new to the team, requires SELECT permission on this table. Through which part of the Databricks Lakehouse Platform UI can the data engineer grant this privilege?

A. Repos

B. Compute

C. Data Explorer

D. Workflows

E. Workspace

You are tasked with implementing data governance policies across multiple business units within a Databricks workspace. Describe how you would use the segregation of business units across the catalog to enforce these policies. Additionally, explain the benefits of this approach in terms of data isolation, access control, and compliance.

A. Segregate business units by creating separate catalogs for each unit, enhancing data isolation and access control.

B. Segregate business units by creating shared catalogs for each unit, reducing data isolation and access control.

C. Segregate business units by creating a single catalog for all units, improving data isolation and access control.

D. Segregate business units by creating overlapping catalogs for each unit, weakening data isolation and access control.

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Which statement about a managed table in Databricks is **incorrect**?

A. Databricks manages both the metadata and the data for a managed table

B. Data for a managed table does not reside in the `LOCATION` of the database it is registered to

C. Managed tables are the default when creating a table in Databricks

D. Dropping a managed table will delete both the metadata and underlying data of that table

E. Managed relationship between the data location and the database means that in order to move a managed table to a new database, you must rewrite all data to the new location

A data engineer is running code in a Databricks Repo that is cloned from a central Git repository. A colleague of the data engineer informs them that changes have been made and synced to the central Git repository. The data engineer now needs to sync their Databricks Repo to get the changes from the central Git repository.
Which Git operation does the data engineer need to run to accomplish this task?

A. Clone
B. Pull
C. Merge
D. Push

In a scenario where you are working with a large dataset in Azure Databricks that contains 'product_name' and 'category' columns, your task is to ensure data integrity by validating that each 'product_name' is associated with only one unique 'category' value. Considering the need for efficiency and accuracy in a production environment, which of the following Spark SQL queries would you use to identify any 'product_name' that violates this uniqueness constraint by being associated with more than one 'category'? Choose the best option.

A. SELECT product_name, COUNT(DISTINCT category) as category_count FROM dataset GROUP BY product_name HAVING category_count > 1

B. SELECT product_name, MAX(category) as max_category FROM dataset GROUP BY product_name

C. SELECT product_name, category FROM dataset GROUP BY product_name, category HAVING COUNT(*) > 1

D. SELECT product_name, category FROM dataset WHERE product_name IN (SELECT product_name FROM dataset GROUP BY product_name HAVING COUNT(DISTINCT category) > 1)

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Which of the following commands can be used to query a delta table in Databricks?

A %sql Select * from table_name

B %python spark.sql("select * from table_name")

C %python execute.sql("select * from table")

D Both A & B

E %python delta.sql("select * from table")

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A data engineer is maintaining a data pipeline. Upon data ingestion, the data engineer notices that the source data is starting to have a lower level of quality. The data engineer would like to automate the process of monitoring the quality level. Which of the following tools can the data engineer use to solve this problem?

A. Unity Catalog

B. Data Explorer

C. Delta Lake

D. Delta Live Tables

E. Auto Loader

A junior data engineer typically uses the `INSERT INTO` command to write data into a Delta table. A senior data engineer recommends a different command to prevent the insertion of duplicate records. Which command does the senior data engineer suggest?

A. APPLY CHANGES INTO

B. MERGE INTO

C. COPY INTO

D. UPDATE

E. INSERT OR OVERWRITE

You are tasked with querying a three-layer namespace in Databricks to retrieve data from a specific table. The namespace consists of catalogs, databases, and tables. Describe the SQL query you would use to access a table named 'customer_info' located in the 'sales' database within the 'enterprise' catalog. Additionally, explain the benefits of using a three-layer namespace in terms of data organization and access control.

A SELECT * FROM enterprise.sales.customer_info; Benefits include hierarchical organization and granular access control.

B SELECT * FROM sales.enterprise.customer_info; Benefits include flat organization and broad access control.

C SELECT * FROM customer_info; Benefits include simplicity and ease of use.

D SELECT * FROM enterprise_sales_customer_info; Benefits include consolidated naming and simplified access.

Where is the Databricks web application hosted according to the Databricks Lakehouse architecture?

A Databricks Filesystem

B Control plane

C Customer Cloud Account

D Data plane

E Databricks-managed cluster

A data engineering team operates a DLT pipeline that updates all tables at specified intervals until it is manually stopped. Upon stopping, the compute resources are terminated. Which execution mode best describes this DLT pipeline?

A. The DLT pipeline operates in Continuous Pipeline mode under Development mode.

B. The DLT pipeline operates in Triggered Pipeline mode under Development mode.

C. The DLT pipeline operates in Continuous Pipeline mode under Production mode.

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If a data engineer does not have the necessary permissions and is unaware of the owner of 'new_table', how can they determine who owns it?

A. Examine the Owner field on the table's page in the cloud storage solution

B. There is no possible way to find out who owns the table

C. Check the Owner field on the table's page in Data Explorer

D. All the provided methods can help identify the table's owner

E. Look at the Permissions tab on the table's page in Data Explorer

The data engineering team is working with a Delta table named `products` that includes product details, such as the net price. Which of the following code blocks correctly applies a 50% discount to all products priced above 1000 and updates the table with the new prices?

A. SELECT price * 0.5 AS new_price FROM products WHERE price > 1000;

B. MERGE INTO products WHERE price > 1000 WHEN MATCHED UPDATE price = price * 0.5;

C. UPDATE products SET price = price * 0.5 WHERE price > 1000;

D. MERGE INTO products WHERE price < 1000 WHEN MATCHED UPDATE price = price * 0.5;

E. UPDATE products SET price = price * 0.5 WHERE price >= 1000;

What major drawbacks are job clusters subject to compared to serverless clusters.

A Limited data governance

B Start-up time

C Higher operational cost

D Lack of Python support